TECHNICAL CASE STUDY

Mineral Prospectivity Mapping from Satellite Imagery

Binary segmentation of mineral-bearing zones from Landsat-8 multispectral data over Salem District, Tamil Nadu — comparing three deep learning architectures under extreme class imbalance.

- **github.com/anitkumarmaity1-hash/Minerals_Prospective_Mapping**

95.11%	92.4%	2,509:1	3
BEST TEST F1-SCORE	BEST TEST RECALL	BACKGROUND-TO-MINERAL RATIO	ARCHITECTURES COMPARED

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U-NET · ATTENTION U-NET · RESUNET

01 — THE PROBLEM

Mineral zones are geologically rare and spectrally noisy

01 **Spectral overlap.** Band ratios and PCA alone can't separate mineral signatures from background rock and soil.

02 **No spatial context.** Pixel-wise statistical methods ignore neighbouring-pixel relationships that CNNs capture naturally.

03 **Severe imbalance.** Verified mineral occurrences cover a tiny fraction of the study area — most segmentation losses collapse to predicting "background" everywhere.

04 **Field surveys are expensive.** A reliable prospectivity map narrows the search area before geologists ever visit the site.

SALEM DISTRICT · VALID PIXELS

5,794,980 valid pixels

Landsat-8, 30m, cropped to boundary

2,309 labelled mineral pixels 0.04%

2,509:1

background-to-mineral pixel ratio — drives every loss-weighting and augmentation decision downstream

02 — PIPELINE

One pipeline, five feature configurations, three models

1. **Acquisition**
   Landsat-8 L2 imagery + GADM, lithology & fault shapefiles
2. **Preprocessing**
   Radiometric scaling, QA cloud masking, boundary crop
3. **Feature Engineering**
   7 raw bands, top-3 PCA components, 5 geological rasters
4. **Ground Truth**
   Mineral points → 90m buffer → rasterised binary mask
5. **Patching & Aug.**
   128×128 patches, stride 64, zero-overlap 70/15/15 split
6. **Training**
   Weighted BCE + Dice, AdamW, early stopping on val IoU

7ch
Raw surface reflectance bands
B1–B7, per-band Z-score normalised

3ch
PCA-derived components
97.68% of spectral variance retained

5ch
Geological rasters
Lithology, intrusive, age, group, distance-to-fault

15ch
Max combined input
Five configurations tested across three models

Best configuration per model, held-out test set

MODEL	BEST INPUT / MECHANISM	TEST IOU	TEST F1	TEST RECALL
U-Net	3-channel PCA input, deep supervision on two auxiliary decoder heads	90.67%	95.11%	92.33%
Attention U-Net	8-channel PCA + geological rasters, soft attention gates on every skip connection	88.84%	94.09%	90.25%
ResUNet ★	15-channel full feature stack, residual blocks with identity shortcuts	88.51%	93.90%	92.42%

Loss: Weighted BCE (pos_weight = 45) + Dice, equal-weighted · **Optimiser:** AdamW, ReduceLROnPlateau · **Stopping:** early stopping on validation Mineral IoU, patience 15

04 — WHY RECALL, NOT JUST ACCURACY

A missed deposit costs more than a false alarm

Under a **2,509:1** background-to-mineral imbalance, IoU and F1 are precision-dominated — they penalise a false positive and a false negative equally. In mineral exploration, that trade-off is backwards.

A false positive costs a short field visit. A false negative means a real deposit is **never investigated** — a permanent, irreversible loss.

ResUNet was selected as the operational model: highest recall (**92.42%**) on the richest 15-channel input, with residual connections preventing gradient degradation across the deeper, heterogeneous feature stack.

U-NET · HIGHEST IOU

90.67%

Best on paper — but 8 more missed mineral pixels than ResUNet on the test set

RESUNET · CHOSEN MODEL

92.42%

Highest recall — maximises mineral detection completeness for field survey planning

Full report, notebooks & code
github.com/anitkumarmaity1-hash/Minerals_Prospective_Mapping